

VEDANT TENDULKAR

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Summary

Final-year Computer Science student with backend and infrastructure experience — FastAPI services, PostgreSQL/vector data layers, and containerized deployments on AWS/Docker — plus performance-critical systems work (sub-30ms inference, SSE/WebSocket streaming). Actively building with AI coding agents as part of the daily workflow, not just as a novelty.

Education

VIT Bhopal University

Sep. 2023 – June 2027

B.Tech Computer Science, Cloud Computing and Automation

Achievements

Won **25,000** and the **"Most Enthusiastic Presenter"** award at Online Innovation Summit 6 for pitching a startup idea to real investors alongside a co-founder.

Experience

Clever Harvey

Business Development Intern

Remote

- Analyzed market trends and customer data to identify partnership opportunities, contributing to a 30% increase in qualified leads.
- Translated technical product concepts for clients and fed insights back to the product team at the product/customer interface.
- Completed a Junior MBA; awarded a [Letter of Recommendation](#) for performance.

Projects

MindMate | *Python, FastAPI, PostgreSQL, pgvector, Next.js, Docker*

[github](#)

- Developed a mental health chatbot featuring four distinct LLM personas via OpenRouter, utilizing pgvector in PostgreSQL to give the bots long-term conversational memory.
- Built the complete backend from scratch, handling JWT authentication, database migrations, and a custom crisis detection system, all backed by 130+ Pytest cases and deployed via Docker.

Skincare Vision Agent | *Python, FastAPI, YOLOv8, ONNX Runtime, WebSockets*

[github](#)

- Trained a custom YOLOv8 model on the ACNE04 dataset to detect skin conditions on the fly, using ONNX Runtime to bring inference time down to under 30 milliseconds.
- Set up a FastAPI server using WebSockets to stream live video predictions, wiring it up to an LLM to instantly generate personalized skincare advice based on the scan.

TurboQuant | *Python, PyTorch, Transformers, Model Quantization*

[github](#)

- Ran extensive benchmarks to test how much LLM text quality drops during model compression, measuring the exact perplexity jumps from full precision (40.48) down to 4-bit and 3-bit formats.
- Wrote custom implementations for KV-cache quantization techniques (like QJL and PolarQuant) based on a recent Google Research paper to cut down GPT-2's memory usage.

EpistemicAgent | *Python, Multi-Agent LLM, ChromaDB (Vector Databases), Groq, RAG*

- Built a multi-agent system where a 'Reader' and a 'Mood Regulator' communicate to process news feeds, using ChromaDB to build memory and dynamically shift their topic interests over time.
- Set up a full RAG workflow using 8-bit quantized embeddings to save space, successfully deploying the agents on Groq to run 70+ autonomous reasoning loops.

Technical Skills

Languages: Python, C++, SQL, JavaScript

Cloud and DevOps: AWS (EC2, VPC, S3, Elastic Beanstalk), GCP, Docker, Vercel, Render, Git, CI/CD

Backend & Frameworks: FastAPI, REST APIs, Object-Oriented Programming

Frontend: React, Next.js, HTML

Databases: PostgreSQL, MySQL, MongoDB, ChromaDB (Vector Databases)

AI Tooling: LangChain, LangGraph, MCP(Model Context Protocol), OpenRouter, PyTorch, ONNX Runtime, YOLOv8, RAG Pipelines